Fine-Tuning Diffusion Models for Few-Step Posterior Sampling by Unfolding and Distillation

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Overview

- Conditional Diffusion Models
- Unfolded Diffusion with LoRA fine-tuning
- Numerical Results

Diffusion Models

Ornstein-Uhlenbeck Process:

$$\mathrm{d}X_t = -\frac{\beta_t}{2} X_t \mathrm{d}t + \sqrt{\beta_t}\mathrm{d}W_t$$

transports $X_0 \sim p(x)$ to $\mathcal{N}(0, 1)$ as $t \to \infty$.
The reverse-time SDE

$$\mathrm{d}X_t = \beta_t \left(\frac{1}{2} X_t + \nabla_x \log p_t(X_t) \right) \mathrm{d}t + \sqrt{\beta_t}\mathrm{d}W_t,$$

transports $X_T \approx \mathcal{N}(0, \mathbb{I})$ to $X_0 \sim p(x)$.

- Approximate X_0 using discrete SDE + Tweedie's identity ($\nabla_x \log p_t(\cdot) \approx -\frac{\epsilon_\theta(\cdot;t)}{\sqrt{1-\bar{\alpha}_t}}$ where $\bar{\alpha}_t = \prod_{s=0}^{t}(1-\beta_t)$).
- Distilled diffusion models (Consistency models, diffusion GANs) reduce the number of discrete SDE steps.

Conditional Diffusion Models

Given degraded observation

$$y = Ax + \mathcal{N}(0, \sigma_y^2),$$

The reverse-time SDE with conditional score

$$\mathrm{d}X_t = -\beta_t \left(\frac{1}{2} X_t + \nabla_x \log p_t(X_t \mid y) \right) \mathrm{d}t + \sqrt{\beta_t}\, \mathrm{d}W_t,$$

transports $X_T \approx \mathcal{N}(0, \mathbb{I})$ to $X_0 \sim p(x \mid y)$.

Tweedie's identity:

$$\nabla_x \log p_t(X_t \mid y) \approx \frac{X_t - \mathrm{E}[X_0 \mid y, X_t]}{2\sigma_t^2}.$$

Conditional Denoising Step

At iteration t, we want to steer samples in the direction of the conditional expectation $\hat{x}_0 = \mathrm{E}[X_0 \,|\, y, X_t]$.
Formulation as a sub-optimization problem:

$$\hat{x}_0 \approx \arg\min_x \log p(x_t, y \,|\, x) + \log p(x)$$

- Data-consistency enforced through proximal gradient step for $p(x_t, y \,|\, x)$.
- Prior modeled implicitly through Plug-and-play denoiser

$$x \mapsto x - \sigma_t \epsilon_\theta(x, t),$$

where ϵ_θ is the score function from a foundational diffusion model.

Unfolded Denoising step

Algorithm Unrolled Sampling from $p(x_0 \mid y, x_t)$.

Require: y, x_t, $A^\dagger y$, Noise Schedule $\{\sigma_t\}_{t=1}^T$
Require: Unfolded Weights θ_{Unfolded}

$\rho_t \leftarrow \gamma \sigma_y \sigma_t (\sigma_y^2 + \sigma_t^2)^{-1/2}$ $\{\gamma > 0$ is a tuning parameter$\}$
$\tau \leftarrow \arg\min_s |\sigma_s - \rho_t|$ $\{$Auxiliary time variable$\}$
$\tilde{x}_{t,0} \leftarrow A^\dagger y$ $\{$Initialize sampler$\}$
for $k = 0, \ldots, K-1$ **do**
  $\hat{x}_{t,k} \leftarrow \text{Prox}_{\rho_t \log p(y, x_t \mid \cdot)}(\tilde{x}_{t.k})$ $\{$Proximal data consistency step$\}$
  $z_{t,k} \sim p_\tau(X_\tau \mid X_0 = \hat{x}_k)$ $\{$Auxiliary noise step$\}$
  $\tilde{x}_{t,k+1} \leftarrow z_k - \sigma_\tau \epsilon_{\theta_{\text{Unfolded}}}(z_k; \tau)$ $\{$Auxiliary denoise step$\}$
end for
$\tilde{x}_{t,K} \approx x_0$ given x_t, y

Low-Rank Fine-Tuning (LoRA)

To avoid expensive memory constraints, the finetuned score model $\epsilon_{\theta_{\text{Unfolded}}}$ uses weights

$$\theta_{\text{Unfolded}} = \underbrace{\theta}_{\substack{\text{Pretrained}\\ \text{Frozen}}} + \underbrace{\color{red}\theta_{\text{LoRA}}}_{\text{Trainable}},$$

$$\color{red}\theta_{\text{LoRA}} \color{black} = AB,$$

where

$$\theta \in \mathbb{R}^N,$$

$$B \in \mathbb{R}^{n \times N}$$

$$A \in \mathbb{R}^{N \times n}$$

for $n \ll N$.

Training

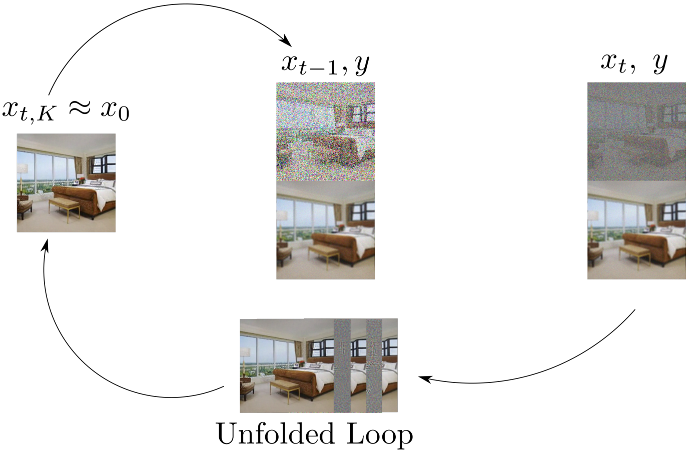
To improve performance over a small number of diffusion steps, we unfold $K = 3$ iterations using a training loss composed of data-consistency, perceptual quality and adversarial metrics:

$$\mathcal{L}(\theta) = \mathrm{E}_{\substack{t \sim \mathcal{U}[[0,T]] \\ x_0, x_t}} \Big[\mathcal{L}_{\mathrm{Adv}}(x_0, \tilde{x}_{t,K}(\theta), D_\phi) \\ + \omega_0 \mathcal{L}_2(x_0, \tilde{x}_{t,K}(\theta)) + \omega_1 \mathrm{LPIPS}(x_0, \tilde{x}_{t,K}(\theta)) \Big],$$

where the discriminator D_ϕ is simultaneously trained to maximize an adversarial loss

$$\mathcal{L}^D(\phi) = \mathrm{E}_{\substack{t \sim \mathcal{U}([[0,T]] \\ x_0, x_t}} \Big[\mathcal{L}_{\mathrm{Adv}}(x_0, \tilde{x}_{t,K}(\theta), D_\phi) + \mathcal{L}_{\mathrm{GP}}(\phi) \Big]$$

Unfolded Diffusion Loop



Inpainting

Pixels masked: 80%, $\sigma_y = 0.05$, $x^\star \in \mathbb{R}^{3\times 256\times 256}$.

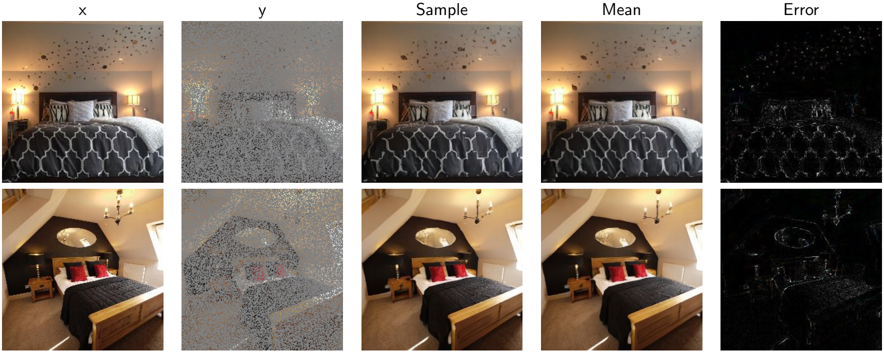


Figure: Inpainting results on LSUN Bedroom

Gaussian Deblurring

kernel size: 25×25, $\sigma_y = 0.05$, $x^\star \in \mathbb{R}^{3\times256\times256}$

(a) $x^\star$ (b) y (c) $\hat{x}_{est}$: $(32.62, 0.07)$ (d) $|x^\star - \hat{x}_{est}|$

Figure: Qualitative results of deblurring on ImageNet.

Gaussian Deblurring

kernel size: 25×25, $\sigma_y = 0.05$, $x^\star \in \mathbb{R}^{3\times256\times256}$

(a) $x^\star$ (b) y (c) $\hat{x}_{est}$: $(29.17, 0.04)$ (d) $|x^\star - \hat{x}_{est}|$

Figure: Qualitative results of deblurring on ImageNet.

Gaussian Deblurring

kernel size: 25×25, $\sigma_y = 0.05$, $x^\star \in \mathbb{R}^{3\times256\times256}$

(a) $x^\star$ (b) y (c) $\hat{x}_{est}$: $(30.33, 0.12)$ (d) $|x^\star - \hat{x}_{est}|$

Figure: Qualitative results of deblurring on ImageNet.

Results

Gaussian deblurring: kernel size 5×5,
Uniform deblurring: kernel size 9×9,
Super-resolution: kernel size 5×5, factor 4
$\sigma_y = 0.01$ (all tasks)

		Deblurring						SR		
		Gaussian			Uniform			$\times 4$		
Methods	NFEs	PSNR	LPIPs	FID	PSNR	LPIPs	FID	PSNR	LPIPs	FID
Ours	9	**38.77**	<u>0.02</u>	<u>4.61</u>	**35.57**	**0.02**	**11.14**	24.42	<u>0.15</u>	20.69
Ours+RAM	9	35.97	**0.01**	**3.30**	-	-	-	**26.70**	**0.08**	**11.9**
CDDB	1000	<u>37.02</u>	0.06	5.01	31.26	0.19	23.15	26.41	0.2	19.88
I2SB	1000	36.01	0.07	5.8	30.75	0.2	23.01	25.22	0.26	24.13
DiffPIR	100	28.10	0.13	21.53	<u>31.44</u>	<u>0.10</u>	20.20	20.39	0.36	70.45
DDRM	20	36.73	0.07	4.34	29.21	0.21	<u>19.97</u>	26.05	0.27	46.49

Table: Results on ImageNet test set. **Bold:** Best result. <u>Underlined:</u> Second best.

Results

128×128 box inpainting,
80% random inpainting,
Gaussian deblurring: kernel size 5×5,
Super-resolution: kernel size 5×5, factor 4
$\sigma_y = 0.05$ (all tasks)

		Inpainting						Deblurring			SR		
		Box			Random			Gaussian			$\times 4$		
Methods	NFEs	PSNR	LPIPs	FID	PSNR	LPIPs	FID	PSNR	LPIPs	FID	PSNR	LPIPs	FID
Ours	9	20.82	0.15	<u>30.3</u>	22.32	**0.19**	<u>43.3</u>	**28.13**	<u>0.12</u>	**23.73**	24.51	<u>0.19</u>	<u>32.71</u>
Ours + RAM	9	20.84	0.15	**16.23**	23.37	<u>0.20</u>	**22.3**	<u>28.03</u>	**0.11**	<u>37.016</u>	25.48	**0.12**	**19.47**
CM4IR	4	-	-	-	**25.28**	0.33	159.43	27.48	0.25	53.19	26.14	0.295	124.22
CoSiGn	2	22.61	<u>0.137</u>	38.64	23.22	0.368	-	19.08	0.54	-	26.13	0.217	40.84
CDDB	1000	**23.74**	**0.126**	45.20	-	-	-	-	-	-	**27.31**	0.24	54.20
I2SB	1000	<u>23.21</u>	0.276	55.10	-	-	-	-	-	-	<u>27.23</u>	0.242	53.40
DiffPIR	100	13.42	0.34	66.79	<u>23.8</u>	0.36	97.69	26.14	0.36	64.81	23.83	0.45	101.92
DDRM	20	18.90	0.223	51.50	-	-	-	27.35	0.28	-	25.09	0.374	-

Table: Results on LSUN Bedroom test set. **Bold:** Best result. <u>Underlined:</u> Second best.

Conclusion

- Algorithm unrolling provides a natural, model-based approach to conditional score evaluation.
- Low-rank perturbations to a foundational score model can be used for lightweight model finetuning with negligible storage requirements.
- Training the conditional diffusion to minimize a mixture of perceptual and data-consistency losses allows for few-step diffusion sampling of the posterior distribution.